

BHAVESH ANANDPARA

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EXPERIENCE

Associate Software Engineer at Bentley Systems

06/2024 - Current

- Engineered a **LangGraph-based agentic workflow** integrating **15+ MCP-enabled engineering tools**, implementing tool routing, stateful execution, and multi-step orchestration to automate complex drawing-production workflows through natural-language instructions.
- Engineered a **sandboxed Code Mode execution framework** that enabled AI Chat feature to generate and execute code against exposed product APIs, using **Skill files, controlled function exposure, and execution guardrails** to automate complex engineering workflows.
- Built an AI-assisted **GitHub PR reviewer recommendation POC** using **Qdrant, embeddings, and vector similarity search** to match changed files with historical developer contributions and automatically rank the **top 1-2 most relevant reviewers**.
- Adopted an **AI-first software development workflow** using **Claude Code, context engineering, reusable Skills, and task-specific agents** for codebase exploration, implementation, pre-PR reviews, architectural validation, and automated test generation while preserving human-in-the-loop quality control.
- Built and evaluated a **VGG16 transfer-learning model** for bridge detection from satellite imagery, achieving **89.7% validation accuracy** and approximately **0.90 precision, recall, and F1-score** across the evaluation set.
- Automated parallel-road mapping across multiple ETL data sources using a **Python/QGIS plugin**, reducing manual database-update effort by approximately **30%** and preventing an estimated **90%+ of cross-source data inconsistencies** caused by manual updates.
- Built **3 GitHub Actions workflows** automating security vulnerability remediation, dependency validation, and release-branch synchronization, preventing dependency issues that previously required up to **half a day** of debugging and remediation.

DevOps Intern at Coditas Solutions LLP

01/2024 - 06/2024

- Deployed cloud infrastructure using **AWS EC2, S3, VPC, ALB, Auto Scaling, CloudWatch, and SNS**, implementing load balancing, horizontal scaling, monitoring, and event-driven alerting for scalable application environments.
- Built **Jenkins CI/CD pipelines** and containerized applications using **Docker**, automating build and deployment workflows and reducing manual release effort.

SKILLS

Languages	Python, TypeScript, JavaScript (ES6+), C#, Java, HTML, CSS
AI & Agentic Systems	LangGraph, MCP, RAG, Agentic Workflows, Multi-Agent Systems, Tool Calling, Context Engineering, Prompt Chaining, Guardrails, Human-in-the-Loop
LLM & Developer Tools	Claude, OpenAI, Claude Code, GitHub Copilot, Code Mode, Skills
Retrieval & AI Data	Qdrant, Neo4j, Embeddings, Vector Search, Semantic Search
Backend	.NET, Node.js, Express.js, Flask, REST APIs
Frontend	React.js, TypeScript, Redux, Context API, Three.js, Tailwind CSS
Cloud & DevOps	AWS, Docker, Jenkins, GitHub Actions, CI/CD
Testing	Playwright, Jest, Postman
PROJECTS	

CodeAtlas – AI-Powered Software Architecture Intelligence Platform

Personal Project

- Built an **AI-powered software architecture intelligence platform** using React, TypeScript, Node.js, and Express, combining **AST-based static analysis** with LLM reasoning to understand public GitHub repositories and extract code structure, dependencies, routes, and relationships.
- Architected a modular AI backend following **Clean Architecture, SOLID, and dependency inversion**, separating repository analysis, dependency graphs, semantic retrieval, documentation generation, and AI reasoning into independent services.
- Built a **hybrid RAG architecture** using **Neo4j knowledge graphs and Qdrant vector search**, combining structural code relationships with semantic retrieval to generate repository-aware documentation, Mermaid diagrams, semantic search, and dependency impact analysis.
- Designed a multi-provider LLM abstraction supporting **Ollama, OpenAI, Claude, and Gemini**, enabling model switching without modifying downstream application logic and supporting flexible AI orchestration across the platform.

EDUCATION

Bachelor of Technology, in Computer Engineering

Bajaj Institute of Technology, Wardha.

2024

8.57 CGPA